Time as a Quantum Memory

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Abstract: In the framework of Presentism, we introduce a novel interpretation of time as a quantum memory, which evolves in atomic instants, and show its compatibility with relativistic time dilations. The scope is to provide additional evidence for minimizing the ontological existence of the time dimension, possibly contributing to a better understanding of its quantum nature. First, we propose our postulates on time, causality, and information, and define our ontology in terms of entanglement in a fuzzy Euclidean lattice encoded in the Present memory. Then, we introduce our observer: an elementary massive particle, described in terms of entanglement in the lattice and across the instants. Finally, we derive the proper time of such particle from the sampling of information in its causal cone between events of collapse. In the context of a universe emerging from information, we conclude highlighting the relevance of the concept of memory in the emergence of complexity.

Keywords: time; presentism; relativity; information; memory; entanglement.

1. Introduction

The understanding of the quantum nature of spacetime has gained interest in recent years, following the search for a theory of Quantum Gravity (QG) able to describe, in a single framework, General Relativity (GR) and Quantum Mechanics (QM). At the core of the problem, as reminded by Smolin in Ref. [1], there is an old question still unanswered: what is time? Recently, in the search for a better understanding of the foundational aspects of our universe, entanglement and information have also been promoted as key elements in the emergence of spacetime, as shown in Refs. [2] and [3].

In this paper, we propose a novel description of the emergence of spacetime inspired by Presentism and Information Theory. Presentism is an ancient interpretation of time for which only the present instant exists, already suggested by Heraclit of Efes and Plato and recently rediscovered by several scholars in physics, including Aharonov *et al.* in Ref. [4].

What seems to be missing, in the literature inspired by Presentism, is a clear indication of how a relativistic description of time intervals could emerge, if only the Present exists. Answers to this open question provide evidence for minimizing the ontological existence of the time dimension, contributing to the foundational understanding of its nature.

We support this debate with a novel interpretation of the Present as an ontological quantum memory and providing a bottom-up derivation of the phenomenology of time dilations. Our work is inspired by the time-symmetric description of the Present of Ref. [4], the insights of Refs. [5] and [6] on the discreteness of evolution and information, and the holographic models of the universe such as Refs. [7], [8], [9], and [10].

To introduce our framework and show its compatibility with the emergence of a proper relativistic experience of time, we proceed as follows:

- In section 2, we briefly present the rich literature on Presentism in physics, mathematics, and philosophy.
- In section 3, we describe the main postulates of Presentism on time and causality and add an element of novelty to the established framework: an explicit relation with holographic information. We describe the Present as an ontological quantum memory, which encodes the information potential in a fuzzy relational lattice and evolves in atomic instants. We conclude with a postulate on the holographic limit of the information in the causal cone from an event of collapse in the lattice.

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- In section 4, we clarify our ontology. We describe the information in the Present in terms of non-locality and undefined causality, intended as entanglement in the lattice and across the instants respectively, and modelled through time-symmetric closed paths in the Present (*memory-loops*).
- In section 5, we introduce our elementary observer: a massive particle. We describe fermions in terms of momenta in space and in time, connected to their non-locality and half spin and interpreted through the memory-loops of section 4.
- In section 6, we combine all the introduced elements and overcome the critics raised against Presentism in terms of compatibility with Special Relativity (SR). We derive the proper time of our particle-observer from the information in the causal cone between events of collapse postulated in section 3.
- In section 7, we conclude suggesting a relation between the concept of *memory* and the emergence of complexity from information at different levels of abstraction.

As a last note, to support the reader, the list of the used symbols is available in the Annex at the end of the paper.

2. Presentism

Interpretations of time inspired by Presentism have now their tradition, established by several scholars of physics, mathematics, and philosophy. In this section, we report a brief list of the main concepts behind this rich framework. We note that, in the literature, the present instant is often also called the *Becoming*, to highlight the idea of transition from what is *undetermined* and *quantum*, to what is *determined* and *classical*.

Smolin highlights the need for a new understanding of time already in Ref. [1], and proposes a description of evolution as a sum of *"views"* of the past in Ref. [11], extended to a *Quantum Mechanics of the Present* in Ref. [12]. Gisin, in Refs. [5] and [6], connects Presentism to intuitionist mathematics, where Real numbers are not "*given all at once*" with infinite information, but "*bit by bit*", in an increasing information instant after instant.

The research of Operational Probabilistic Theories (OPT), which starts from Information Theory to interpret QM and has been introduced in Ref. [13] and [14], is based on discrete operations and circuits (*foliations*) that evolve in atomic computational instants, similar to atomic present instants.

Eliztur, in Ref. [15], speculates on a time symmetric *Becoming* as a bridge between QM and SR. The absence of a preferred arrow of time within an atomic Present has also been proposed by Aharonov *et. al.* in Ref. [4], where each instant is as a "new universe" (inspired by Heraclit). In Aharonov model, unitarity comes from maximal entanglement between subsequent moments, while events of collapse disentangle adjacent instants. Cohen, Cortez, Elitzur, and Smolin, in Ref. [16], focus as well on a time symmetric model of the Present, and extends to *Energetic Causal Set* in the past of the Present in Ref. [17]. Finally, in the framework of time-symmetric models, we investigate the information potential in the Present and its connection with non-locality and undefined causality in Refs. [18] and [19].

Kauffmann, in Refs. [20] and [21], elaborates on the description of the present instant as connected to the Heisenberg *Res Potentia* of QM, different from the classical *Res Extensa* of the irreversible spacetime. Riek, in Refs. [22] and [23], investigates the implications of a discrete evolution and the need for a "thickness" in time to distinguish, in an event, the cause from the effect, while Schlatter, in Refs. [24], [25], and [26], elaborates on the concept of *synchronization* and of a spacetime emerging from irreversible events.

In Presentism, the discreteness of time is usually extended to space. This is proposed in the context of a physically realizable universe, for which the information density required to describe it must be finite. From a mathematical perspective, a finite information cannot describe the Real continuum, as elaborated by Gisin in Ref. [5], and no existing universes should need infinite or unbounded information to be physically representable.

The finiteness of information density, given that information is generally associated with energy, is supported also by the Bekenstein bound of Ref. [27], and leads to a picture of the universe that can be both relativistic and indeterministic in its evolution, very far from the block-universe of *Eternalism*, as illustrated in Ref. [28].

The finiteness of information suggests a possible indeterminacy of the past as well. Recent *gargantuan simulations* reported in Ref. [29] have shown that time seems irreversible at the most fundamental level, beyond a thermodynamic arrow of time: even a 3-body system "*would require an accuracy of smaller than the Planck length in order to produce a time reversible solution*". The irreversibility of events has also been related to the concept of *irretrodicibility*, as illustrated by Del Santo and Gisin in Ref. [30] with a model to calculate *propensities* of past events. In a framework inspired by Presentism, what has already happened can influence the Present but cannot be changed and, beyond fundamental limits, cannot even be known with certainty.

Presentism is a relevant model of time in modern philosophy as well. To support the reader, we list here a few references, which investigate the historical, logical, and metaphysical aspects of this interpretation, such as Refs. [31], [32], [33], [34], [35], [36], [37], [38], [39], [40], [41] and [42]. In the philosophical research on the compatibility with SR, Craig in Refs. [43] and [44] advocates for a Neo-Lorentzian interpretation, but these arguments fail on many counts, as shown in Ref. [45]. Our bottom-up derivation of the relativistic phenomenology from information provides evidence in favor of Presentism and offers insights on the ontological nature of information also on a philosophical ground.

We conclude this short introduction highlighting the relevant role of Presentism in the debate on the human perception of time as well. Concepts such as the *specious present*, defined as the temporal interval in which one's perceptions are considered to be *in the present*, have been investigated from a philosophical, psychological, and neurological perspective (see, for example, Refs. [46] and [47]). We do not elaborate here on these topics, but we believe that evidence in favor of Presentism from a perspective rooted in elementary physics and focused on information could provide insights also on the human perception of time, as briefly suggested in the conclusion.

3. Postulates

Postulate on Time

The main assumption of Presentism is summarized in our first postulate:

Only the Present exists, as an atomic instant of evolution ΔT *of the whole universe* P. 1

Postulate P. 1 implies that evolution in the universe occurs in discrete/thick atomic instants of temporal duration ΔT. This discrete evolution is synchronous in all the universe, as a global "update cycle" that marks the progression of instants as a universal clock.

Each present instant, that we call a universe *tick*, identifies the evolution of space as a foliation. This foliation has a thickness in time of magnitude ΔT, and it is considered equivalent to a circuit foliation of OPT, introduced in Ref. [13] and [14]. In this discrete evolution, we label the Present as the instant T_k. The instant T_{k-1} is considered the immediate past, while T_{k+1} the immediate future. Postulate P. 1 implies that T_{k+1} does not exist yet, while T_{k-1} does not exist anymore.

We call *causal time* the ordered set of instants prior to T_k. This set identifies an oriented and discrete axis of evolution, from a distant past to more recent instants of the universe. The causal time labels the passage of instants as a classical Newtonian time of our universe and does not ontologically exist. Being the present instant T_k the *kth tick* in the discrete evolution of the universe, the concept of the "age of the universe" is then intended as a temporal interval $\Delta t_U = k\Delta T$ from a *first tick* T_1 of the universe.

It is crucial to clarify that the Present is not an *observers' common now*. Observers, as clarified in section 6, can only compare differences in the number of ticks, and an extended causal time axis (to count the passage of time with a proper number of ticks) is not defined at the fundamental level of abstraction on time given by the present instant only.

Postulate P. 1 implies a shortest time interval ΔT, duration of each atom of evolution. This discrete evolution implies a maximum rate of change. In this sense, ΔT is also proposed as the "temporal resolution" of the occurrence of events.

The boundary between T_k and T_{k-1} (defined as the *past boundary* of the Present) separates what has already happened (events) from what is still possible (what has not happened yet, but may happen in the present instant, given the past consequences at the end of T_{k-1}). The Present, as per the referenced literature, *encodes* the possible *becoming* of information from undetermined and potential to determined and classical.

In our framework, events are irreversible as they do not exist anymore, and something that does not exist cannot be acted upon, nor changed. The irreversibility of the past, as well as its limited *retrodicibility*, has been elaborated in Refs. [28], [29], and [30], and it is foundational in a model of time that considers only the Present as ontologically existing. Even if events do not exist in the Present, from a given event at an instant, a causal cone of information propagates as the universe ticks occur, in a sphere growing in each instant. The information in these causal cones is the basis for a relativistic evolution, as shown in section 6. In this sense, events in the causal time influence (shape) the Present.

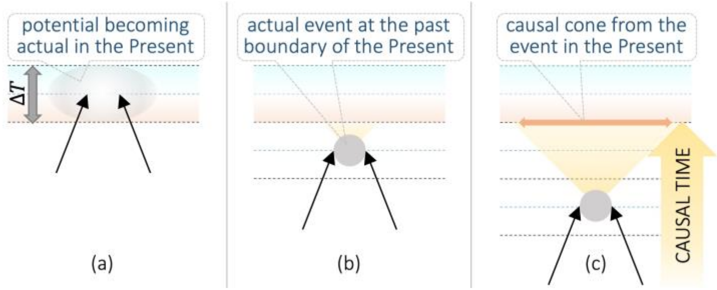


Figure 1. Representation of the *Becoming* in 3 consecutive universe ticks, as 3 "snapshots" (a, b, c). Each snapshot is shown with the Present at the top and the past ticks "sinking" as the causal time grows. In (a), we picture a possible event in the Present from past energy-momenta (*black arrows*), that becomes an actual event (*grey dot*) at the past boundary of the Present in (b), and from which a causal cone propagates (as information in the Present) as the ticks occur in (c).

We clarify that the causal time extends up to the past boundary of the Present, giving causal order to the set of events, but it is not properly defined within the Present. As in Refs. [4], [11], [12], [15], [16], [17], [18], [19], [20], [21], [24], [25], [26] and, as originally suggested by Plato himself (see Ref. [38]), in our model the Present is not part of the causal time, nor of a classical and deterministic spacetime.

In the encoding of the *Becoming*, we do not consider a defined arrow of causality within the thickness of the Present, as elaborated in the QM models reported in Refs. [4], [15], [16], [17], [18], and [19]. The absence of a preferred arrow of time has been described, in these models, through a time-symmetric formalism in each instant, with a superposition of a forward and a backward contribution within ΔT. We note that a time-symmetric formalism to model the information within the Present suggests a description of the thickness of each instant as $\Delta T = 2T$. From this equivalence, being the Present intended as the k^{th} tick from the birth of our universe, the label $(2k-1)T$ marks the past boundary of the Present, while $(2k+1)T$ marks the future boundary of this atom of time, that we consider of Planckian nature (T = Planck time).

We conclude on P. 1 considering the following definitions inspired by Ref. [20]:

- *Res Extensa*: what has already happened, intended as the classical domain of events behind the past boundary of the Present, irreversible, and ordered in a causal set along the causal time.
- *Res Potentia*: what could happen in the current instant given the causal past, intended as the evolution from the events' consequences, possible events in the Present and the quantum information encoding what is still undefined.

Given P. 1, the information in the k^{th} tick ontologically exists in the Present. We therefore describe the Present as an ontological quantum memory that encodes the *Res Potentia* and evolves in atomic computational cycles $\Delta T = 2T$. We clarify our ontology based on information and entanglement to describe the *Res Potentia* in section 4, after having concluded our investigation on the main postulates of our framework.

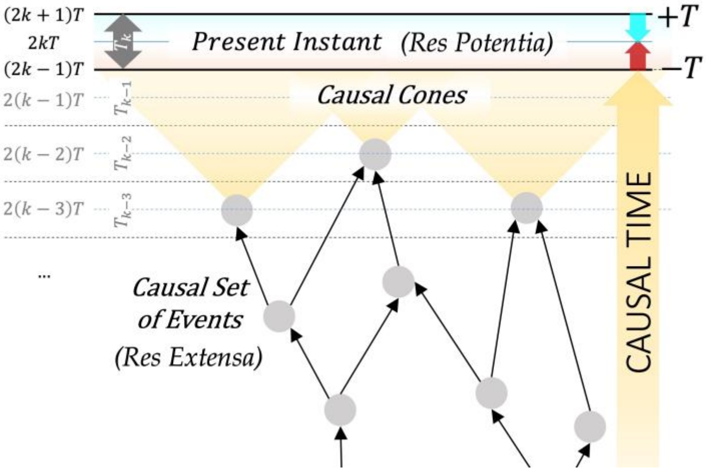


Figure 2. Model of a universe evolving in instants. From the top, we show the Present T_k, which encodes the *Res Potentia* in a time symmetric formalism (*cyan/red arrows*). It is the k^{th} tick from the first instant, between $(2k-1)T$ and $(2k+1)T$. Below, a causal set of events (*grey dots*) connected by energy-momenta (*black arrows*) along an emerging causal time (*yellow arrow*). These events are part of a classical domain of *Res Extensa* and influence the Present through their causal cones.

Postulate on Causality

Descriptions of our universe inspired by Presentism have a strong connection with the concept of causality. An evolution in atomic present instants is naturally related to the information in the causal set behind the past boundary of the Present, as already elaborated by several scholars, including Smolin *et. al.* in Refs. [11], [16], and [17]. The relevance of causality is not limited to the information in the past, as it is also a fundamental aspect in the quantum evolution in each instant, as elaborated by Aharonov *et. al.* in Ref. [4], Elitzur and Dolev in Ref. [15], and Del Santo and Gisin in Refs. [28] and [30]. In this paper, we consider a postulate on causality that focuses on its speed, intended as a maximum rate of change of the possible spatial position over the universe ticks.

The speed of causality c is constant in the whole universe P. 2

In our discrete evolution, we consider a possible event at the present instant, such as the emission of a photon from a particle at a position X at T_k, or simply X_k. From X_k, we describe the other possible events' locations in the same instant through a Wick rotation: we consider an imaginary time it_X to reach these other possible events' locations at the speed of causality c (defined in P. 2), and 2 angular degrees of freedom $\theta_{1,2}$. Given a discrete temporal evolution from P. 1, the imaginary time is intended as discrete as well, and it is defined through multiples of an imaginary atom $i\Delta T$. From a given X_k, we map the other possible events' locations at $n_X \in \mathbb{N}^+$ *imaginary steps* $ic\Delta T$, in an imaginary path $icn_X\Delta T$ from X_k. These imaginary paths, which are related to the possible dynamics of a system (*ala Path Integral*) trace a *fuzzy Euclidean space*, as in Refs. [48], [49], and [50].

We consider the set $\{in, \theta_{1,2}\}_X$ of all possible events' locations from X_k. This set maps to a discrete lattice, which extends in 3 dimensions and with a resolution $|ic\Delta T| = 2L$ (being L = Planck length). Given that we have posed no conditions on X_k, this mapping represents a relational description of a fuzzy Euclidean lattice. We therefore call *imaginary space* or *fuzzy space* the lattice of all possible events' locations. We conclude that, in a Presentism perspective, space at T_k from X_k is fuzzy, Euclidean, relational, and discrete.

We clarify that the imaginary time is orthogonal to both the causal time axis and the symmetric arrows of time in the Present (as per space and time orthogonality). The causal time identifies time-like separated events, while the fuzzy space describes space-like separated possible events, and it is intended as a boundary in the thickness of the Present. Being the Present defined between $(2k-1)T$ and $(2k+1)T$, this fuzzy imaginary space is defined as a *Cauchy surface* at $2kT$, which identifies a past and a future bulk in the thickness of the Present, from $(2k-1)T$ to $2kT$ and from $(2k+1)T$ to $2kT$ respectively.

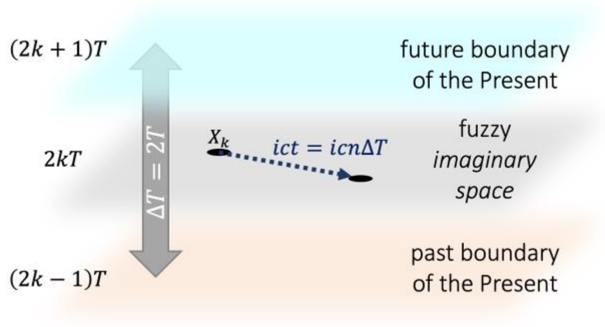


Figure 3. Representation of the fuzzy space within the Present instant. We show the past and the future boundaries (*red/cyan*) and 2 locations in the fuzzy space separated by an imaginary path $icn\Delta T$, which is defined through a Wick rotation.

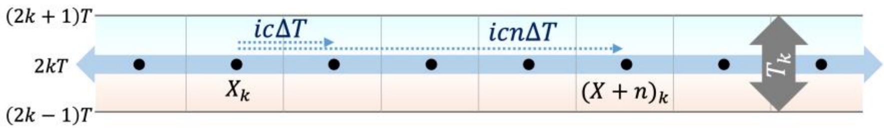


Figure 4. Discrete set $\{in, \theta_{1,2}\}_X$ from an atom of space X (with $\theta_{1,2}$ omitted for clarity). We show the imaginary space in the Present as a lattice (black dots). It is defined, with a Wick rotation, through an imaginary time counting the steps $ic\Delta T$, in a relational description of the fuzzy space of possible events. We also show the past and the future bulk in the Present with shades of red/cyan.

Postulate on Information

We conclude section 3 adding an explicit relation with information. While the first two postulates are commonly accepted in the literature inspired by Presentism, although with differences in each description, this postulate is something we add here to the picture.

We first remind the *Nyquist–Shannon sampling theorem* of Ref. [51]: a continuous signal with a limited bandwidth can be equivalent to the information in its discrete samples. Being a universe evolving in atomic computational instants ΔT equivalent to a signal with maximum bandwidth $1/\Delta T$, according to this theorem, a continuous spacetime experience can emerge from its discrete samples.

We also clarify that we attribute to this information a physical nature, as proposed by Landauer in Ref. [52], and discourage the idea that we live in a simulation. The atomic tick ΔT (rate of change in time) and the imaginary step $ic\Delta T$ (rate of change in the fuzzy space) give physicality to our discrete representation. Global references for physical rates of change seem needed not only to define any meaningful comparison of spacetime intervals between independent observers, but also to allow a coherent structure between far regions of spacetime, as in a universal encoding. In our model, these references define the basis through which temporal intervals and spatial distances emerge from the abstract entities of discrete and finite information.

Finally, we introduce our last postulate on the *information in the causal cone* $\varsigma_{e,N}$, which is related to the information rate as the ticks occur. Given N instants from the last collapse of a particle at a location X, we define the information $\varsigma_{e,N}$ as the number of angular Degrees of Freedom (DoF) on the sphere of radius $cN\Delta T$ from X. This information can be understood as an "increasing resolution" of the surface of a Bloch sphere centered in X, in a saturation of the Bekenstein bound of Ref. [27].

The information $\varsigma_{e,N}$ is related to the *entanglement entropy* of Ref. [53] and it is equivalent to the entropic information of Refs. [9] and [10]. Specifically, it can be described in terms of entropy if we consider the surface of the causal cone after N ticks as a horizon at N steps of $\frac{Area}{(c\Delta T)^2} = 4\pi N^2$ possibles DoF. The set of all entangled configurations between the 2 sides of this horizon sums to $W = 2^{4\pi N^2}$, with an entropy $log_2 W = \varsigma_{e,N}$. We postpone the detailed study of the relation between our framework and Refs. [9] and [10] to a future work, and consider $\varsigma_{e,N}$ as a postulate in our model:

The information $\varsigma_{e,N}$ in the causal cone of N instants
saturates on the surface of a sphere of radius N: $\varsigma_{e,N} = 4\pi N^2$ P. 3

We suggest that the particle "samples" $\varsigma_{e,N}$ in terms of an increasing non-locality in the fuzzy imaginary space and undefined causality across the instants (given the absence of causal stimuli from the last collapse that would have caused a new event of collapse). The information of non-locality $\varsigma_{s,N}$ and of undefined causality $\varsigma_{t,N}$ sampled by the particle are both intended in terms of angular DoF from X after N instants.

Given N_s imaginary steps in an undefined direction $\theta_{1,2}$ from X, and N_t ticks experienced by the particle with no new events (therefore in an undefined causality), we consider the following decomposition of the information in the causal cone $\varsigma_{e,N}$:

$$\varsigma_{e,N} = 4\pi N^2 = \varsigma_{t,N} + \varsigma_{s,N} = 4\pi N_t^2 + 4\pi N_s^2 \tag{1}$$

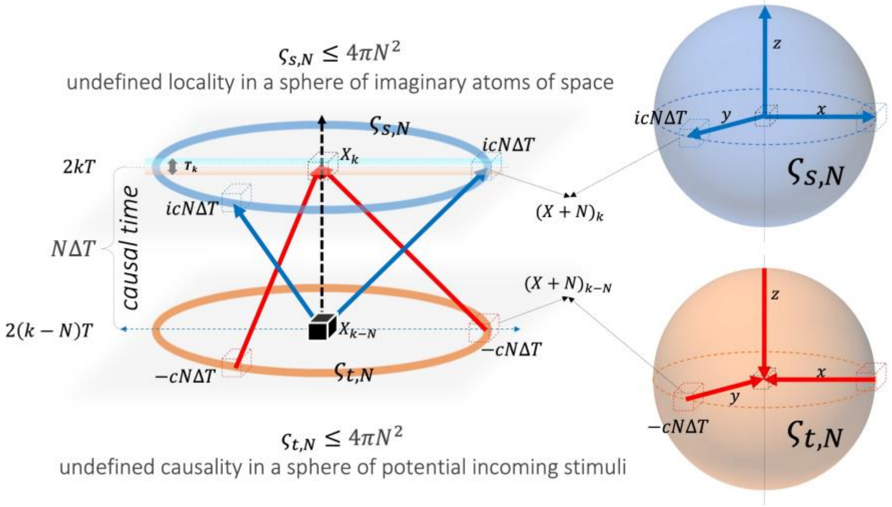


Figure 5. Max information of non-locality $\varsigma_{s,N}$ and undefined causality $\varsigma_{t,N}$ after N ticks from a collapse at an atom X. On the left, we show the cones, with k marking the ticks on the vertical axis. On the right, we show the 3D fuzzy space with k implicit. *Blue* figures represent the max $\varsigma_{s,N}$ that could be sampled, while *red* figures the max $\varsigma_{t,N}$ that could be sampled.

4. Ontology

In section 3, we have described the Present as an ontological memory encoding the potential in a fuzzy lattice and evolving in atomic instants. Moreover, we have introduced our postulates on the rate of change in time, in space, and of information in the causal cone, setting the basis for a relativistic evolution. In this section, we clarify how the *Res Potentia* of possible events in the fuzzy lattice is encoded in the memory of the Present.

We are interested in encoding the undefined information of possible events' location (*the particle is at X xor Y*) and possible events' order across the instants (*the particle was first at A and then at B xor first B and then A*). This information, having no over/under-determined solutions, is undefined but logically consistent, as shown in Refs. [18] and [54].

To encode this information, we highlight the relation between causal non-separability and cyclicity of the causal structure illustrated in Refs [55] and [56], and that no events, nor a defined causal orientation, can be identified on a closed time-like curve (CTC), as clarified in Refs. [57], [58] and [59]. We also note that loops are the most basic circuits to implement a memory (see Ref. [60]).

As already proposed in Refs. [18] and [19], in the context of the time-symmetric description of the *Res Potentia* suggested by Aharonov in Ref. [4], we therefore consider time-symmetric possible paths which superpose in a loop as the ideal *circuit* to encode the undefined information of possible events' location (non-locality) and possible events' order (undefined causality). We call this closed path a *memory-loop*, as it acts as a *storage* able to encode a still undefined information.

To clarify our description of the information in the *Res Potentia*, we first elaborate on the non-locality in the fuzzy space. According to Aharonov in Ref. [4], non-locality in the k^{th} instant is encoded in a pair of Hilbert spaces, $\mathcal{H}_k$ and $\mathcal{H}_k^\dagger$, in a dual cover of the space of possibilities at $2kT$ from the boundaries $(2k\!-\!1)T$ and $(2k\!+\!1)T$. In Aharonov model, $\mathcal{H}_k$ encodes *ket* vectors $|\psi\rangle$, while $\mathcal{H}_k^\dagger$ is mapped to *bra* vectors $\langle\psi|$, with a max correlation $|\psi\rangle\langle\psi|$ between instants and disentanglement in case of an event $\langle\psi|\psi\rangle$ at a given instant. In our time-symmetric description, as per Ref. [4], $|\psi\rangle$ is the *initial* (*forward*) wave from $(2k\!-\!1)T$ to $2kT$, while $\langle\psi|$ is the *final* (*backward*) wave from $(2k\!+\!1)T$ to $2kT$.

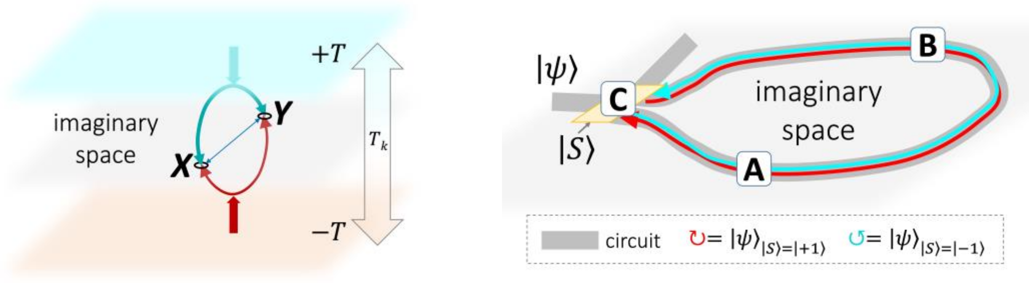
In the extension of Ref. [4] in [19], a non-local potential between 2 possible locations, X and Y, is encoded in the symmetric bulks of the Present $\mathcal{H}_k$ and $\mathcal{H}_k^\dagger$ through the superposition of time symmetric paths from *(2k–1)T* and *(2k+1)T*. These paths meet at $2kT$ as a loop in the thickness of the Present, "piercing" the imaginary space in X and Y. Along this loop, given that no events occur on a CTC (as per Refs. [58] and [59]), a particle in the Present is non-locally in X *xor* Y (Fig. 6a). Through this loop in the thickness of T_k, these possible events' locations are eventually connected by a tunneling potential *ala* $ER = EPR$ conjecture, proposed in Refs. [7] and [8], and extended to $ER = EPR = CTC$ in Ref. [18].

Besides the entanglement of possible events' location, we consider the entanglement in the time order of possible events. Entanglement in time has been studied with equivalent Bell's inequalities and experimentally verified in Refs. [61], [62], [63], [64], and [65]. These references describe a *quantum-SWITCH* circuit able to superpose, through a controller qubit $|S\rangle$, the order of possible events: $\{A \to B\} \oplus \{B \to A\}$ (being A and B two distinct locations along the possible path of a photon in a circuit).

As elaborated in Ref. [18], this information can be encoded in a closed imaginary path, traced in the fuzzy space as the ticks occur. This closed path comes from the superposition of a path in which A is met first and B last, with a path that met B and then A (Fig. 6b). The superposition of these paths is described, in our time-symmetric formalism, as a forward and backward propagation along the imaginary time superposed thanks to the qubit $|S\rangle$ in a closed loop in the fuzzy space: $\big(|\circlearrowright\rangle_{|S\rangle=|+1\rangle}\colon\{A \to B\}\big) xor \big(|\circlearrowleft\rangle_{|S\rangle=|-1\rangle}\colon\{B \to A\}\big)$. This loop in the fuzzy space encodes a state entangled in time:

$$|\circlearrowright\rangle_{|S\rangle=|+1\rangle} \oplus |\circlearrowleft\rangle_{|S\rangle=|-1\rangle} = |\odot\rangle_{|S\rangle} \tag{2}$$

In short, in our description of entanglement as an ontological information encoded in the Present as memory-loops, a *loop in time* (orthogonal to the fuzzy space, Fig. 6a) encodes entanglement in the fuzzy space (undefined locality of possible events), while a *loop in space* (developing as imaginary closed path orthogonal to the Present thickness, Fig. 6b) encodes entanglement in time (undefined order of possible events).



(a) (b)

Figure 6. (a) Entanglement in space as a superposition of a forward (*red*) and a backward (*cyan*) wave from $\pm T$ closed in a loop in the Present. This loop pierces the imaginary space in X and Y, connecting them (*ala ER=EPR=CTC*). (b) Entanglement in time as a superposition of imaginary paths of a particle $|\psi\rangle$ controlled by $|S\rangle$. The undefined order $\{A \to B\} \oplus \{B \to A\}$ is encoded in an imaginary closed path in the fuzzy space between the controller $C_{|S\rangle}$ and any two locations, A and B, on the circuit.

We highlight that both kinds of loop encode the causal non-separability of possible events through a cyclicity of their causal structure, as per Refs. [18], [55] and [56]. This is a possible description of how, according to entropic theories (see Ref. [9]), "*space is a storage*" (implemented, in our model, through its thickness in time). We also clarify that, as long as there is a closed path, no events occur (Refs. [58] and [59]), and the information is encoded as still undefined. The opening of a memory-loop, on the other hand, is a post-selection of a path that has causally occurred. It is a disentanglement which leads to a new irreversible event in the causal structure behind the past boundary of the Present and an extension of the particle causal path (word-line) which emerges along the causal time (*e.g.:N ticks ago, the particle was at the atom* X). With the opening of a memory-loop, we therefore follow the same interpretation of the collapse suggested in the time-symmetric description of disentanglement and event of Ref. [4].

5. Observers

Momenta in Space and Time

In this section, we introduce our elementary observer: a massive particle, described through the memory-loops of section 4, and as per the toy-model introduced in Ref. [19]. The concept of a "particle-observer" is introduced to represent an elementary physical entity with a subjective physical experience of time aligned with Einstein's proper time.

We start from a description of the particle's wave function in terms of momenta in space and time inspired by Dirac, then elaborate on its non-locality encoded in the Present in the context of the time-symmetric description of Ref. [4] and the holographic perspective of Ref. [9], and conclude suggesting a possible relation between the half spin of fermions, their invariant mass, and an entanglement across the instants.

To describe a particle in the Present, we consider a decomposition of the energy in orthogonal components (*Energy-momentum* relation introduced by Dirac, also known as *Einstein triangle*). Being m_T the invariant mass, $m_r = m_T/\sqrt{1-\beta_r{}^2} = \gamma m_T$, and $v_S = \beta_r c$ the translational velocity of the Center of Momentum (CoM), we define the *momentum in time* p_T and the *momentum in space* p_S:

$$\begin{cases} p_T = m_T c \\ p_S = m_r v_S = (\gamma m_T)(\beta_r c) \end{cases} \tag{3}$$

The momentum p_S is a vector with 3 imaginary components and origin at the Center of Momentum (CoM) of the particle (as the imaginary part of quaternions). The momentum p_T represents the invariant information in time and it is defined in the thickness of the Present, orthogonal to all the 3 imaginary dimensions of space. We define $p_{ST} = p_T + ip_S$, which is related to the energy by Eq. (4) (from here on, we consider in our notation a single imaginary direction ict, usually intended as the direction of the velocity v_s).

$$E^2 = |p_T c|^2 + |ip_S c|^2 = |p_{ST} c|^2 = (p_{ST} c)(p_{ST}^* c) \tag{4}$$

Being $\alpha_r = 1/\gamma = \sqrt{1-\beta_r{}^2}$ and m equivalent to the mass m_T in Planck units:

$$p_{ST} = p_T + ip_S = \frac{\hbar}{cT}\gamma m(\alpha_r + i\beta_r) = \frac{\hbar}{cT}\gamma m \varepsilon_r = |p_{ST}|\varepsilon_r; \qquad |\varepsilon_r| = 1 \tag{5}$$

To show the relation between p_{ST} and the non-local potential of the particle in $icn\Delta T$, we consider the wave function $\psi(n)$ from the CoM of the particle at an atom O at T_k:

$$\psi(n) = \frac{R(n)}{cT} e^{\frac{iS(n)}{\hbar}}$$

We consider the gradient ∇ from O in the imaginary distance axis ict as $\nabla = \frac{i\partial}{\partial icT} = \frac{\partial}{cT}$, with ∂ intended as a variation in the imaginary space from O along a single imaginary direction for simplicity but, in general, ∇ is intended as applied to a vector with 3 components, of which $|ir|$ is the magnitude of the distance in the imaginary space.

In line with a model of *zitterbewegung* with max entropy, we describe $R(n)$ as shaped in a thermal distribution from O, function of a tunneling amplitude β_m, which is related to the momentum in time and the reduced Compton wavelength $\frac{i\hbar}{p_T} = icn_c T$ as per Eq. (6):

$$|\psi(n)| = R(n) = \frac{p_T c}{\hbar/T} e^{-\frac{p_T c}{\hbar/nT}} = \frac{1}{n_c} e^{-\frac{n}{n_c}} = \beta_m e^{-\beta_m n} \tag{6}$$

Besides the momentum in time, the momentum in space p_S is equivalent, as usual in QM, to the gradient of the phase S of ψ. We can therefore express p_T and p_S also in terms of ψ:

$$\begin{cases} p_T = -\hbar\nabla \ln R = -\left(\frac{\hbar}{cT}\right)\partial \ln R \\ ip_S = i\nabla S = \frac{i\partial S}{cT} \end{cases} \tag{7}$$

In short, p_T encodes the invariant information $\nabla \ln R$, while p_S the relativistic component proportional to ∇S. As a synthesis able to relate the momenta of a free fermion (with CoM at *O*) with the scale of its non-locality, we connect p_{ST} and ψ with Eq. (8).

$$\nabla\psi = \nabla\frac{1}{cT}e^{\left(\ln R+\frac{iS}{\hbar}\right)} = \left(\nabla\left(\ln R + i\frac{S}{\hbar}\right)\right)\frac{R}{cT}e^{\frac{iS}{\hbar}} = -\frac{(p_T - ip_S)}{\hbar}\psi = -\frac{p_{ST}^*}{\hbar}\psi$$

$$\nabla\psi = \frac{\partial\psi}{cT} = -\frac{p_{ST}^*}{\hbar}\psi$$

$$\frac{p_{ST}^* c}{\hbar/T} = -\frac{\partial\psi}{\psi} = -\partial\ln\psi \tag{8}$$

Non-Locality from the Boundaries

How should we interpret Eq. (8) and the non-locality of the particle in Presentism? We propose a correspondence of the boundaries of the Present with the limits of the fuzzy space. This description is similar to *Celestial Spheres* (see Refs. [66]) and the models in which the *holographic screen* is the boundary in quantum-to-classical *transactions* (*e.g.:* Ref. [67]), but considers 2 boundaries, $\mp T$, in a time-symmetric framework. Specifically, through a stereographic projection at the atom *O* (where the CoM is located), we map $-T$ to the point at *null* ($ice^{\varphi}T|_{\varphi\to-\infty}$, *inside O*) and $+T$ to the point at *infinity* ($ice^{\varphi}T|_{\varphi\to+\infty}$, *outside O*). The limit at *infinity* is not a physical infinity but a tangent limit of this stereographic projection, while the limit $\varphi \to 0$ in $ice^{\varphi}T$ maps the boundary of the atom *O*, at a radial distance $|icnT|$ from *null*. In this mapping, the thickness of the Present is eventually equivalent to the *coarse graining variable* of holographic theories (direction corresponding to *scale*, see Ref. [9]).

As per Refs. [4] and [19], we then relate $p_{ST}c$ with the *ket* $|\psi\rangle$ in $\mathcal{H}$ (*forward* wave), while p_{ST}^*c to the *bra* $\langle\psi|$ in $\mathcal{H}^\dagger$ (*backward* wave, as in Wick rotated models, complex conjugation equals time-reversal). In this time-symmetric description, $|\psi\rangle\langle\psi|$ encodes the non-local potential with a dual cover from *null* ($|\psi\rangle$ *ex-ante*) which is retro-closed from *infinity* ($\langle\psi|$ *ex-post*, up to where the particle could tunnel at a given instant). We clarify that, in the decomposition of p_{ST} in its orthogonal components, p_T encodes the non-local potential as a vector from $\mp T$, while ip_S encodes the variation of the CoM across the ticks.

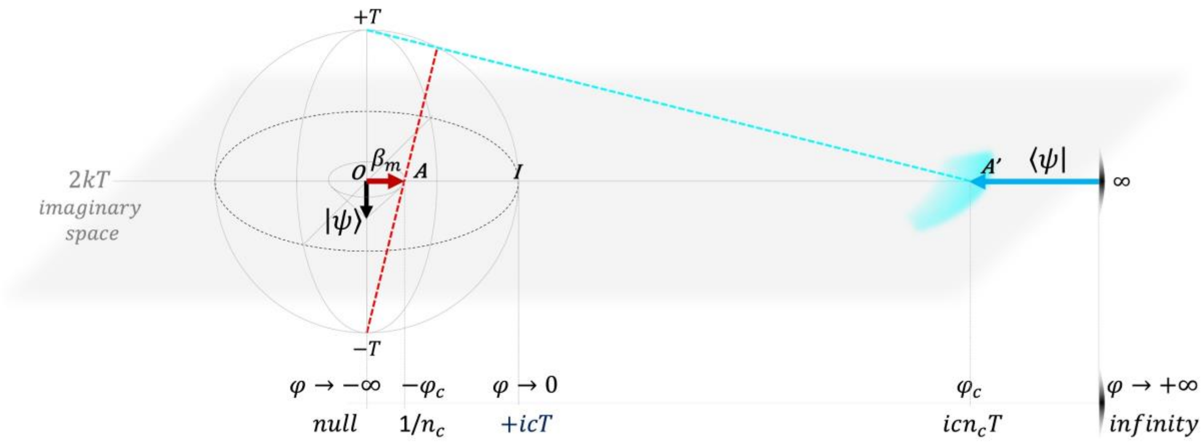


Figure 7. Fermion with CoM at the atom *O*, depicted through a stereographic projection from $-T$ and $+T$ (*red/cyan* dashed lines) mapped to *null-infinity* in the imaginary space. The atom of space *O* is shown as a disk on the hyperplane of the imaginary space. We also illustrate $|\psi\rangle$ (*black arrow*), the amplitude β_m of Eq. (6) (*red arrow*), and the final wave $\langle\psi|$ retro-closed from *infinity* (*cyan arrow*).

In terms of memory-loops, the non-locality encoded in $|\psi\rangle\langle\psi|$ is described through bundles of loops in the thickness of the Present. Each bundle of loops pierces the fuzzy space in a sphere of imaginary locations from *null* with a probability as per the Born rule. Being $\sum_n 4\pi n^2|\psi(n)|^2 = Z$, we can write:

$$\wp\{collapse@(O+n)\} = \wp\{\langle\psi(n)|\psi(n)\rangle\} = 4\pi n^2|\psi(n)|^2/Z \tag{9}$$

We clarify that an event of collapse $\langle\psi|\psi\rangle$ at a given tick and location, as per Ref. [4], is a disentanglement between successive instants (post-selection of a single actual path) and it is intended as the "point particle". Being an event, which belongs to the irreversible *Res Extensa*, the point particle does not ontologically exist in our framework.

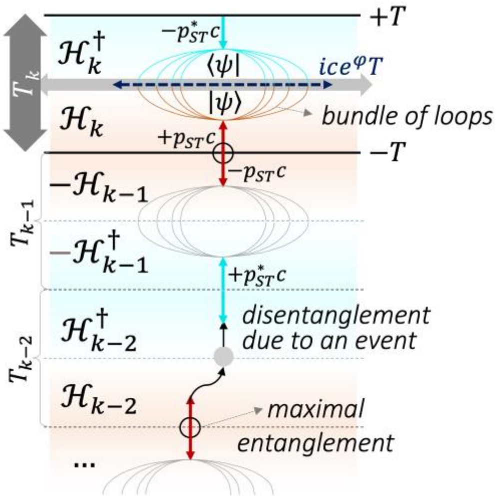


Figure 8. Representation of the evolution of a particle in the last few instants up to the Present. We picture the imaginary space axis in T_k as $ice^{\varphi}T$. The energy $p_{ST}c$ and $p_{ST}^{*}c$ are shown as vectors from the boundaries of the Present, as per Aharonov in Ref. [4] (*red* in $\mathcal{H}$, *cyan* in $\mathcal{H}^{\dagger}$). They are at the root of the initial $|\psi\rangle$ and final $\langle\psi|$ wave closed between $\pm T$ in a bundle of loops. Max entanglement among successive instants (*black circle*) is broken in case of an event (*e.g.*, *grey dot* at the instant T_{k-2}). We remind that only the present instant exists; the previous ones are shown for illustrative purpose.

Fermionic Spin and Entanglement in Time

A description of particles from entanglement in space is not new the literature, as shown in Refs. [2] and [3]. We believe that a model based on Presentism may also offer useful insights towards a description of the rest mass as an information *persisting in time* and related to an entanglement across instants.

To elaborate on this, we consider the fermionic spin. In Ref. [19], a connection between spin ½ and an extended period of revolution of magnitude $2\Delta T = 4T$ has been discussed. This relation has already been considered in theories with a discrete time (*e.g.*, in Ref. [68], a fast oscillation between 2 states is proposed as a deterministic model of spin). In our interpretation, we see the relation between spin ½ and an extended period of revolution in terms of universe ticks as a phenomenon to be interpret in terms of entanglement in time.

First, we remind that a particle revolves in the fuzzy space from both *ex-ante/inside* ($-T$ as *null*) and *ex-post/outside* ($+T$ as *infinity*). These 2 sides of the boundary of the atom O (where the CoM is located) are mapped to the double-cover of the fuzzy space in $\mathcal{H}$ and $\mathcal{H}^{\dagger}$ identified by Aharonov in Ref. [4], with the backward paths $\langle\psi|$ acting as *strings from infinity* which retro-close the non-local potential in the fuzzy space (Fig. 7, cyan arrow).

In this representation, we can describe the revolution of the fermion across the instants as a Dirac spinor following the metaphor of the *belt in the book* used by Penrose in Ref. [69]. Specifically, the *strings from infinity* $\langle\psi|$ can be interpreted as the *belt in hour hand* of this metaphor, with the other extreme anchored on the *book-particle* (on the surface of the imaginary sphere where the particle could emerge). Given a 2π revolution in each instant, the particle therefore needs 2 ticks for the 4π revolution required to *untwine the belt* around its CoM, complete its period, and return to its original state.

The fermion, to complete this 4π revolution in $2\Delta T$, could spin clockwise or counterclockwise, with a consistency on the direction of revolution between successive ticks according to a qubit $|S\rangle$, that is encoded on the boundaries of the Present (which act as controller in this circuit-revolution across instants). $|S\rangle = |+1\rangle$ maps, for example, to a clockwise 4π revolution $|\circlearrowright\rangle_{|S\rangle=|+1\rangle}$ in 2 ticks, while $|S\rangle = |-1\rangle$ to the opposite $|\circlearrowleft\rangle_{|S\rangle=|-1\rangle}$. The superposition of these paths closes in a loop in the imaginary space and encodes an entanglement in time, as per section 4. We describe this superposition as a memory-loop controlled by $|S\rangle$ with the same formalism of Eq. (2): $|\circlearrowright\rangle_{|S\rangle=|+1\rangle} \oplus |\circlearrowleft\rangle_{|S\rangle=|-1\rangle} = |\odot\rangle_{|S\rangle}$.

These imaginary paths trace loops in the fuzzy space of length statistically equal to the Compton wavelength, as their radius from *null* depends on β_m, which shapes $|\psi|$ as per Eq. (6). The center of the loops in the fuzzy space identifies the CoM of this persisting information, while the undefined spinning direction across the instants is encoded in $|S\rangle$.

Concluding, the spin ½ of fermions is, according to our proposal, strictly related with an entanglement in time and with the nature of fermions as particles with an invariant mass and a CoM possibly *at rest* in space. Specifically, we suggest that the momentum p_T, which encodes the rest mass as a vector orthogonal to the fuzzy space, emerges from the entanglement in time of these imaginary closed paths, traced by the tunneling amplitude β_m in its undefined spinning in the fuzzy space.

To our knowledge, in the context of a Presentism framework, the connection between spin ½, entanglement in time described as a closed path in a fuzzy space, and the rest mass, is a distinctive feature of our proposal but, beyond the Presentism setting, these ideas are not new in the physics literature. To support our description of the particle-observer on a theoretical ground, we therefore highlight the study on the principles needed to ensure the consistency of observers in terms of information reported in Ref. [70]. In this contribution, the author shows that each *observer is linked with internal self-consistent loops,* similar to our time-symmetric closed path proposed to describe the spin of fermions. Moreover, the relation between the half spin of fermions, a *time-loop* (similar to our loop along the imaginary time), and a *time-directed screw dislocation vector* (similar to our p_T) has also been elaborated in Refs. [71] and [72] in the context of a parallel between QG models and defects in Dirac-materials. The time-loops of these works measure the magnitude of the defect, as our momentum p_T encodes the scale of the information at rest in space. This analogy may represent a promising theoretical and experimental path connected to our framework, to be investigated in future contributions.

As a final note on our elementary observer, we clarify that a fermion, through its entanglement in time, implements a memory of its state at the past instant. We remind that, in our model, only the Present ontologically exists, and that this memory is intended as the capability of a fermion to discriminate the existence of successive instants while entangling in time (spinning in its imaginary loops $|\odot\rangle_{|S\rangle}$ over 2 ticks), in a proper experience of the passage of time. It does not imply the concurrent existence of several instants.

6. Spacetime

The Relativity of Information Sampling

In section 5, we have described a particle through its momentum in time and in space. We have connected the momentum in time to its intrinsic non-locality and an entanglement across the instants, while the momentum in space to the variation of the CoM in the fuzzy space, in an increasing non-locality. In this section, we show how these momenta are connected to the information introduced in P. 3 and that, thanks to Eq. (1), the concept of a present instant is compatible with SR and the emergence of relativistic time intervals.

Following the "*classes of emergence*" of Ref. [3], we also take the opportunity to clarify that the emergence of spacetime from information is intended as *Hierarchical*, similarly to several holographic models.

To show how spacetime emerges for the elementary observer of section 5 and how its proper time is derived, we start by considering a fermion that last collapsed at a position X at T_{k-N}, N instants before the Present, with $N \gg \frac{\hbar/T}{p_T c}$. The information $\varsigma_{e,N}$ in the causal cone, according to P. 3, is equal to $4\pi N^2$ angular DoF. Given Eq. (1), a fermion "samples" $\varsigma_{e,N}$ through an entanglement in space and in time from the collapse. The information $\varsigma_{s,N}$ is sampled as increased non-locality in the fuzzy space related to the momentum in space p_S. The information $\varsigma_{t,N}$ is sampled as increased undefined causality from the last collapse (while spinning in its imaginary loops, given the absence of incoming causal stimuli that would have caused a new collapse) and it is related to the momentum in time p_T. These relations are described by the following equations:

$$\begin{cases} \dfrac{\varsigma_{t,N}}{\varsigma_{e,N}} = \left(\dfrac{N_t}{N}\right)^2 = \dfrac{|p_T|^2}{|p_{ST}|^2} = {\alpha_r}^2 \\ \dfrac{\varsigma_{s,N}}{\varsigma_{e,N}} = \left(\dfrac{N_s}{N}\right)^2 = \dfrac{|ip_S|^2}{|p_{ST}|^2} = {\beta_r}^2 \end{cases} \tag{10}$$

From Eq. (10), we can rewrite Eq. (1) as follows:

$$\varsigma_{e,N} = \varsigma_{t,N} + \varsigma_{s,N} = 4\pi {N_t}^2 + 4\pi {N_s}^2 = 4\pi(\alpha_r N)^2 + 4\pi(\beta_r N)^2 = 4\pi N^2 \quad (11)$$

In Eq. (11), $\varsigma_{t,N}$ is intended as a sphere of undefined causality of radius $-\alpha_r cN\Delta T$ from X_k, which represents the particle proper experience of the passage of ticks: a temporal interval $\Delta\tau = N_t\Delta T$ from the last collapse. Similarly, $\varsigma_{s,N}$ represents the imaginary surface of radius $+i\beta_r cN\Delta T$ (imaginary path of N_s steps in an undefined direction from *X*) and identifies the set of locations where the particle could quantum jump at the N^{th} instant.

In our discrete evolution, the proper time depends on the reduced information sampled through entanglement in time, given that part of the total information in the causal cone from the last collapse is sampled in an increased non-locality. The proper time therefore emerges from a subsampling of the universe ticks, and it is calculated as the fraction of the causal time experienced by the particle between its events of collapse.

Given a causal time interval $\Delta t = N\Delta T$, a particle sampling $\varsigma_{s,N} = 4\pi(\beta_r N)^2$ in the fuzzy space along the *N* ticks then samples, in the same instants, $\varsigma_{t,N} = \varsigma_{e,N} - \varsigma_{s,N}$ through entanglement in time. In case of a collapse event at the N^{th} tick, the particle has then experienced a proper number of ticks equal to N_t, which is equivalent to a proper time interval $\Delta\tau$ as per Eq. (12), in line with SR result:

$$\Delta\tau = N_t\Delta T = \sqrt{\frac{\varsigma_{t,N}}{\varsigma_{e,N}}}N\Delta T = \sqrt{\frac{\varsigma_{e,N} - \varsigma_{s,N}}{\varsigma_{e,N}}}\Delta t = \sqrt{1 - {\beta_r}^2}\Delta t \quad (12)$$

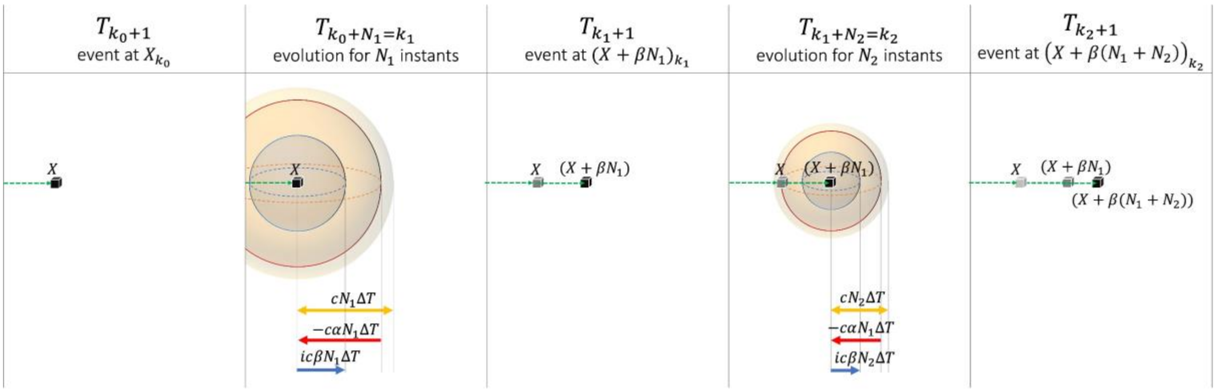


Figure 9. Evolution of a particle between 3 events. Each panel represents a snapshot at a given tick. We highlight the atom of space where the particle is located at the different collapse events (*cubes*), the particle's world-line (*green dashed arrows*), the information sampled between the events of collapse (*blue and red shaded spheres*), as well as the information $\varsigma_{e,N}$ (*yellow shaded spheres*). We note that events and world-lines do not exist in the Present and are shown in the figure for illustrative purpose.

We clarify that, from the last collapse event, we are not allowed to consider an "average uniform motion till the Present", as no movement is causally defined until a new event of collapse occurs. In an evolution based on a discrete time and tunneling events, there is no uniform motion, but an average speed emerging from a series of collapses.

If, at the N^{th} tick from the last collapse, there is a new collapse event, the information of undefined causality and non-locality in the causal cone is released: the particle causally evolves by $N_t\Delta T$ (quantum jump in time of its internal clock) and its CoM results relationally displaced from *X* by $|icN_s\Delta T|$ (quantum jump in space), with an average classical speed between collapses of magnitude $v_s = \beta_r c$.

Extremes and Consequences

To further clarify the proposed model, we elaborate on the most extreme scenarios, related to the sampling of information through entanglement in space or in time only.

We consider the scenario $\varsigma_{e,N} = \varsigma_{s,N}$. It is straightforward to show that such particle experiences no proper time: it evolves counting steps and never samples ticks. In this paper, we do not consider such particles as properly defined observers, as there is no spacetime emerging from their perspective, but the present instant only.

A bath of such particles, such as the photons in the Cosmic Microwave Background (CMB), can represent a statistical reference to define average velocities, as experimentally verified in Refs. [73] and [74] through the study of the CMB dipole.

We consider the scenario $\varsigma_{e,N} = \varsigma_{t,N}$. We suppose that the origin of every particle in the universe occurred with some initial kinetic energy. Moreover, even if in a discrete evolution we can consider a potential resting condition in an instant, there is no chance for an observer to be in such state for long. Given that there are no observers deterministically at rest that could serve as "absolute clocks" to measure the whole set of universe ticks, then it is only possible to compare relative differences in respect to other observers/clocks. Observers can only verify proper relativistic experiences of time intervals, comparing their proper counting of ticks. In any case, for every observer, a proper subsampling of the universe ticks is expected even without other observers to confront with.

From Eq. (12), we also deduce there is no "common now" nor absolute simultaneity for observers. A single universe tick is the same "common now" only for colliding particles in the instant of the event becoming actual (as in Fig. 1a). From the perspective of a single observer, we can only consider a "proper local now", related to the proper experience of the surrounding universe, and an "extended proper present", connected to the observer's proper number of ticks experienced between collapses.

From Eq. (12), we can describe the phenomenology of time dilations. According to SR, a clock with an internal period Δt_0 when at rest, has a period $\Delta t_r > \Delta t_0$ when moving at average speed $v_s = \beta_r c$ in respect to its resting position. The same time interval measured as N_0 periods of duration Δt_0 by the stationary clock, will be measured with only N_r "stretched proper periods" of duration Δt_r when part of the information between collapses $\varsigma_{e,N}$ is sampled as a non-local potential. Mathematically: $N_r \Delta t_r = N_0 \Delta t_0$, with $N_0 > N_r$.

The non-locality between collapses of the moving clock is balanced by a reduced sampling of information through entanglement in time. This leads to a reduced proper rate of change. The dilated period Δt_r (slower frequency) depends on the smaller number of universe ticks counted by the moving clock over the ones causally passing in the universe.

If a full cycle at rest needs N universe ticks ($\Delta t_0 = N\Delta T$), according to Eq. (12), in the same N ticks, the moving clock has experienced $N_t = \alpha_r N$ proper ticks. To complete its cycle, the moving clock needs N proper ticks, and this happens only after N/α_r ticks in the causal time. The "dilated period" Δt_r of the moving clock can then be calculated as:

$$\Delta t_r = \frac{N}{\alpha_r}\Delta T = \frac{1}{\sqrt{1-{\beta_r}^2}}\Delta t_0 = \gamma \Delta t_0 \tag{13}$$

which confirms, as promised result of this paper, the compatibility of a model inspired by Presentism with the relativistic phenomenology of time dilations.

7. Conclusion

We have proposed a framework inspired by Presentism and shown its compatibility with relativistic time dilations. This results leverage on the time-symmetric description of the potential in the Present of Ref. [4], builds on the intuitions of Ref. [6] on the need for a progressive discrete evolution, and extends the Presentism paradigm with a holographic description of the ontological information in each present instant, supported by Ref. [9].

The Present has been described as an ontological quantum memory, which evolves in atomic instants and encodes the still undefined information in a fuzzy Euclidean lattice. We have described the non-locality in the lattice and the undefined causality across the instants in terms of entanglement, modeled trough the superposition of symmetric possible paths closed in loops, and related collapse events with the opening of such loops.

We have introduced our elementary particle-observer as a persisting information locally encoded in the fuzzy space through entanglement across the instants and derived its proper time from the DoF entangled in time between collapses. In our description, fermions experience the passage of time between collapse events as they integrate (sample) entanglement information along an elementary differential step (the atomic evolution cycle of the memory T_k).

Concluding thanks to an interpretation of Presentism through the lens of Information, we can consider the emergence of a (special) relativistic experience of time from the existence of a single instant. This is relevant from a foundational understanding of the quantum nature of time, as it provides evidence for minimizing the ontological existence of the time dimension. If a relativistic experience of time can emerge from a single instant, given as the only one existing, why should we consider the past as existing as well?

We note that a more complex observer than a fermion might encode an extended interval of causal time thanks to the complexity of its interconnected parts, and model in its internal components "more spacetime" than just the intervals between collapses. However, at the most fundamental level, more memory than a thick atomic instant seems not needed.

Finally, we take the opportunity to conclude with a brief digression beyond physics and highlight the relevance of the concept of *memory* in several fields related to the emergence of complexity.

In our model, as in other holographic theories, everything emerges from information and a memory to encode this information seems therefore fundamental. In Biology, DNA is often described as a chemical memory encoding the data needed for life. Moreover, besides the connection between memory, the human perception of the passage of time and of the self-identity (Locke, in Ref. [75]), in recent studies of cognitive neurology (Ref. [47]), also awareness and consciousness have been connected to the access to a *real time memory* (still active while sleeping, but suppressed during anesthesia). The deep relation between memory and consciousness has also been investigated in the context of quantum information in Refs. [76], [77], [78], and [79].

Beyond elementary and biological memories, we also note that the creation of novel content in Generative AI models, even if very far from a conscious experience, depends on a kind of memory. *Transformers*, as clarified in Ref. [80], increase the coherence in patterns recognition and creation, endowing the network with *attention* layers, which eventually act as a memory for non-local correlations in the data.

The emergence of more complex types of *memory* to encode the relevant data at a given level of abstraction on information seems to be a key element to understand the emergence of complexity, from the most fundamental possible *physical layer* of our universe, such as the Present, up to the most complex entity we could discuss about, such as consciousness. As extreme synthesis of the relation between the concept of memory and the emergence of complexity, from the existence of our universe to the awareness of our existence, we therefore dare to formulate:

$$memoro\ ergo\ sum$$

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Annex - Symbols

We list here, by order of introduction, the main symbols used in this paper. We also provide the cross-reference to the section, postulate, or equation where it is first defined.

ΔT	Present temporal extension (*thickness*). See P. 1 Universe *tick* (atomic instant of evolution of duration $\Delta T = 2T$)
T	Planck Time
T_k	The Present (k^{th} instant, after a time of $2kT \pm T$ from the first tick) Also intended as the ontological memory encoding the information potential of the present instant (section 3)
$ic\Delta T$	Imaginary step $ic2T = i2L$ in the fuzzy Euclidean space. See P. 2
L	Planck Length
$\varsigma_{e,N}$	Information in the causal cone. See P. 3 and Eq. (1)
$\varsigma_{s,N}$	Information of undefined locality in the fuzzy space. See Eq. (1)
$\varsigma_{t,N}$	Information of undefined causality across the instants. See Eq. (1)
N_s	Number of imaginary steps between collapses. See Eq. (1) and (10)
N_t	Proper number of ticks between collapses. See Eq. (1) and (10)
$\lvert\circlearrowright\rangle$ *and* $\lvert\circlearrowleft\rangle$	Forward and backward imaginary paths. See Eq. (2)
$\lvert\odot\rangle$	Entanglement in time as a closed path in the fuzzy space. See Eq. (2)
m_T	Invariant (rest) mass. See Eq. (3)
m_r	Relativistic mass. See Eq. (3)
v_s	Translational velocity of the CoM. See Eq. (3)
p_T	Momentum in time. See Eq. (3)
ip_S	Momentum in space. See Eq. (3)
p_{ST}	Momentum in spacetime. See Eq. (4)
β_r	Translational velocity over the speed of causality. See Eq. (5)
α_r	Inverse of the Gamma factor. See Eq. (5) Also intended as subsampling of universe ticks. See Eq. (12)
m	Invariant mass in Planck units. See Eq. (5)
n_c	Reduced Compton wavelength in Planck units. See Eq. (6)
β_m	Fermion intrinsic/invariant tunneling amplitude. See Eq. (6)
ψ	Wave function of a free fermion from its CoM. See Eq. (6)
R	Amplitude of the wave function ψ. See Eq. (6)
S	Phase of the wave function ψ. See Eq. (7)

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